T3. Frequency measurements with ring resonators

Answer sheet

Part A. Field profile in fiber optics (2.5 points)

A1 $C_1 =$ $C_2 =$ **0.3**

A2 $k_{\text{in}} =$ $k_{\text{out}} =$ **0.5**

A3 **0.3**

A4 $B =$ **0.3**

A5 **0.3**

A6 **0.3**

A7 Equation: **0.3**

A8 Equation: **0.2**

Part B. Fiber resonator transmittance (3 points)

B1 Equation: **0.2**

B2 $B_{\text{in}}(t) =$ **0.3**

B3 $B_{\text{in}}(t) =$ **0.3**

B4 $B_{\text{in}}(t) =$ **0.5**

B5 $N_2 =$ **0.5**

B6 Plot: **0.6**

B7 $Q =$ **0.6**

Part C. Calibration of the Mach-Zehnder Interferometer (2 points)

C1 Plot: **0.5**

C2 Plot: **1.0**

C3 $\Delta\omega_{\mathrm{MZI}}/\omega_{\mathrm{MZI}} =$ **0.5**

Part D. Cubic nonlinearity and solitons (2.5 points)

D1 $v_g =$ **0.1**

D2 **0.6**

D3 $D_1 =$ $D_2 =$ **0.5**

D4 Condition on D_2: **0.3**

D5 Plot: **0.6**

D6 $\Delta\omega =$ **0.2**

D7 $\tau_s =$ **0.2**